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Dendrite Growth in Lithium/Polymer Systems

A Propagation Model for Liquid Electrolytes under Galvanostatic Conditions

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Dendrite growth in a parallel-electrode lithium/polymer cell during galvanostatic charging has been modeled. The growth model is surface-energy controlled, incorporating the effect of dendrite tip curvature into its dendrite growth kinetics. Using data representative of the oxymethylene-linked poly(ethylene oxide)/LiTFSI system, it is shown that dendrites accelerate across cells under all conditions, and that growth is always slowed by lowering the current density. Cell shorting occurs during typical charges at current densities above 75% of the limiting current. Increased interelectrode distance slows failure, but the advantages decrease as distance lengthens. A factor of 1000 increase in surface forces delays cell failure by only 6%. While larger diffusion coefficients usually extend the time to cell failure, this trend is not consistent at high transference numbers.
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Lithium batteries were proposed more than 30 years ago as an ideal system for high energy-density applications. In the early 1970s, organic electrolytes were discovered to form passivating films that chemically protect pure lithium electrodes over time.[1] Although this solid-electrolyte interphase (SEI) prevents chemical reaction at a state of no current, a number of failure mechanisms have been discovered that occur during cycling of lithium rechargeable batteries. One of these mechanisms is dendritic growth, first hypothesized to occur in the lithium/polymer system in 1974.[2]

Dendritic lithium deposits, sometimes called mossy lithium after being subjected to multiple charges and discharges, were first directly observed in 1980.[3] Since then, scanning electron microscopy has been used to characterize more thoroughly the appearance of lithium dendrites after cell failure.[4] While work on the lithium electrode is recent, empirical observation and theoretical analysis of dendrite growth in other electrochemical systems has been pursued for years. The most significant of these results are presented here and used to prepare a model that simulates lithium dendrite growth under galvanostatic conditions.

The name dendrite derives from the treelike shape of the deposit, which can range from nearly linear and pointed (needle-like) to highly branched (bush-like). Past work has primarily considered the propagation rate and morphology of dendrites.[5] Most early experiments rely on microscopy, but qualitative pictures resulting from computer simulation have been presented in the last twenty years. Aogaki[6] simulated the electrode appearance during dendrite formation using a three-dimensional linear stability analysis. Chen and Jorné[7] used a fractal growth model to show that macroscopic shapes and branching can be mimicked by changing the dendrite resistance.

Many references in the current literature show the effect of additives on dendritic deposits. Most of the work on additives tends to be simple characterization, and is relatively system-specific. The goals of these experiments include developing more inert solvent/solute mixtures, using novel anions, and incorporating additives to improve plating morphology. In addition to outlining these goals and citing some recent references, Eweka *et al.*[8] provided a typical qualitative examination of additives for increased cycling efficiency in the lithium/polymer system. Kan *et al.*[9] gave a similar example for a zinc/polyaniline battery.

Data related to experimental conditions suggest macroscopic system variables to incorporate in modeling. Jorné and Lii[10] showed that impinging flow toward the electrode prevents formation of dendrites. Yoshimatsu *et al.*[4] found that increased pressure between the electrodes lowers the dendritic growth rate. The effect of various gelation agents on cycle life was examined by Eichinger and Fabian.[11]

Combined, the qualitative work reveals that dendritic growth is a multifaceted problem. Although the dominant effects leading to dendrite growth are electrochemical, a detailed simulation of dendrite growth may require momentum and heat transfer, polymer properties, and detailed description of interfacial phenomena. In order to develop a plan of attack for modeling, it is useful to survey the theoretical work on lithium dendrites, which dates back to the early 1960s.

Barton and Bockris[12] put forth the first comprehensive model of dendrite growth. Their theory contends that plating is faster on protrusions from electrodes because spherical rather than linear diffusion conditions dominate. Because larger spherical diffusion flux leads to narrowing of the dendrite tip, surface forces must be incorporated to resist infinite dendrite thinning. The model presumes that the kinetics of the electrochemical reaction are linear and that the dendrite is a static hemisphere; inertial and mechanical forces are neglected. The dendrite growth rate is maximized with respect to tip radius. Diggle *et al.*[13] fleshed out the previous theory, allowing for higher overpotentials by using a Butler-Volmer kinetic expression and relaxing the assumption that the dendrite is static. These two models compare well to experimental results for steady-state growth of silver dendrites in well-supported electrolytes.

A Mullins-Sekerka type analysis was used by Aogaki and Makino[14] to analyze the stability of electrodes in diffusion-limited cathodic electrodeposition. They found that the surface roughened for every value of the surface energy, revealing that growth of dendrites cannot be prevented by altering the properties of the interface. A thorough analysis with respect to most of the experimental variables in a parallel-electrode geometry was provided by Sundström and Bark,[15] who incorporated transient concentration and potential profiles into their stability analysis to achieve a similar result. Roha and Landau[16] provided an interesting theoretical treatment of leveling agents in electroplating, showing that additive-modified diffusion can lead to smooth deposits during electrodeposition.

More recently, Chazalviel[17] provided a theory of dendritic growth at high currents and/or potentials. In that case, the rate of dendrite growth was predicted by assuming that the electroneutrality condition is violated in the vicinity of the cathode. The model applies only above the limiting current, and thus is not useful for application to secondary batteries. Yamaki *et al.*[18] confirmed that lithium dendrites are needle-like and analyzed the morphology of lithium growth through cracks in the SEI. However, this analysis makes broad assumptions about the flow behavior of lithium which may not apply.

This empirical and theoretical base provides a great deal of in-

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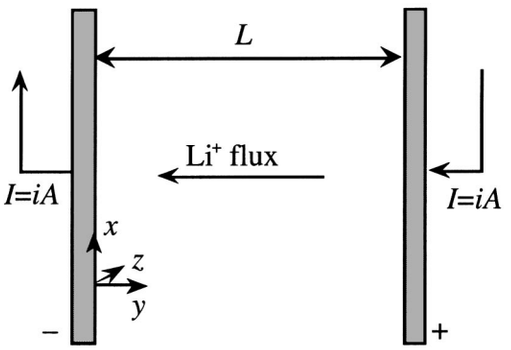


Figure 1. Schematic diagram of the galvanostatic polarization cell.

formation regarding cathodic electrodeposition and many possible directions to take in analysis. In the following sections we describe the transient behavior of a parallel-electrode system and employ some of the concepts discussed above to develop a dendrite growth model. After the theoretical groundwork is developed, we use it to examine growth of dendrites in lithium/polymer cells.

Methods

Concentration profile.—In order to simulate dendrite growth, we first need to know the concentration and potential profile of a binary electrolyte in a galvanostatically cycled planar cell. A typical charge cycle is assumed to last three hours. Figure 1 shows the cell geometry used for this analysis. We assume that the electrolyte behaves ideally, *i.e.*, that the solution is relatively dilute and that the activity coefficients are unity. Material and transport properties are treated as constant with respect to concentration. The cell is tresumed to be isothermal. We treat the system as infinite in the x and z directions, and therefore as one dimensional. The current being passed is I, or, if the electrode area normal to y is A, the current density is $i = I/A$. The interelectrode distance is L.

We wish to find the transient potential and concentration profiles given an initial bulk concentration (c_b) of lithium salt in the separator. If there is no supporting electrolyte, the concentration and potential solutions can be decoupled. Assuming the ions have the same charge number and that electroneutrality holds throughout the cell, we need only to solve the transient diffusion equation for the overall salt concentration (c)

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial y^2} \qquad [1]$$

where the Laplacian has been reduced to one dimension. In Eq. 1, D is the salt diffusion coefficient and t is time. The equation is subject to the boundary conditions

$$i = -\frac{DF}{(1 - t_+^0)}\frac{\partial c}{\partial y}\bigg|_{y=0} \qquad [2]$$

$$\frac{1}{L}\int_0^L c(y,t)dy = c_b \qquad [3]$$

$$c = c_b \text{ at } t = 0 \qquad [4]$$

where F is Faraday's constant and t_+^0 is the cation transference number. Equation 2 indicates that the anion controls the migration of the salt; Eq. 3 expresses conservation of mass. Equations 1-4 can be nondimensionalized according to

$$\xi = \frac{y}{L}$$

$$\tau = \frac{Dt}{L^2}$$

$$\theta = \frac{(c - c_b)DF}{(1 - t_+^0)iL} \qquad [5]$$

which recast our problem as follows, noting that i is negative if the cathode is at 0

$$\frac{\partial \theta}{\partial \tau} = \frac{\partial^2 \theta}{\partial \xi^2} \qquad [6]$$

$$\frac{\partial \theta}{\partial \xi}\bigg|_{\xi=0} = -1 \qquad [7]$$

$$\int_0^1 \theta_T(\tau,\xi)d\xi = 0 \qquad [8]$$

$$\theta = 0 \text{ at } \tau = 0 \qquad [9]$$

First let us analyze the steady-state formulation of these equations, namely

$$0 = \frac{d^2\theta_{SS}}{d\xi^2} \qquad [10]$$

$$\frac{d\theta_{SS}}{d\xi}\bigg|_{\xi=0} = -1 \qquad [11]$$

$$\int_0^1 \theta_{SS}(\xi)d\xi = 0 \qquad [12]$$

The general solution to Eq. 10 is

$$\theta_{SS} = B\xi + C \qquad [13]$$

To satisfy boundary condition 11, we must have that $B = -1$. Boundary condition 12 yields the final steady-state solution

$$\theta_{SS} = -\xi + \frac{1}{2} \qquad [14]$$

In order to solve the transient problem with a Fourier series, we require that the transient profile have homogeneous boundary conditions. This can be achieved by decomposing θ as follows

$$\theta(\xi,\tau) = \theta_{SS}(\xi) + \theta_T(\xi,\tau) \qquad [15]$$

Relation 15 modifies the equation and boundary conditions 6-9 to

$$\frac{\partial \theta_T}{\partial \tau} = \frac{\partial^2 \theta_T}{\partial \xi^2} \qquad [16]$$

$$\frac{\partial \theta_T}{\partial \xi}\bigg|_{\xi=0} = 0 \qquad [17]$$

$$\int_0^1 \theta_T(\tau,\xi)d\xi = 0 \qquad [18]$$

$$\theta_T = -\theta_{SS} \text{ at } \tau = 0 \qquad [19]$$

We can solve this set of equations using separation of variables. Finding Fourier coefficients that satisfy boundary conditions 17-19 yields

$$\theta_T = -4\sum_{k=1}^{\infty} \frac{1}{(2k-1)^2\pi^2} e^{-(2k-1)^2\pi^2\tau}\cos[(2k-1)\pi\xi] \qquad [20]$$

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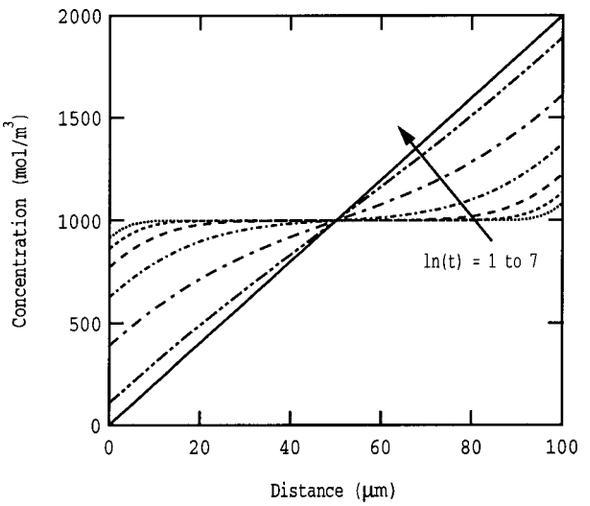


Figure 2. Concentration profile in the galvanostatic cell. In this example, $c_b = 1$ M, $i = i_L$, $D = 5 \times 10^{-12}$ m^2/s, $t_+^0 = 0.3$, and t is in seconds.

The final solution is expressed using Eq. 15, 14, and 20

$$\theta = -\xi + \frac{1}{2} - 4\sum_{k=1}^{\infty} \frac{1}{(2k-1)^2\pi^2} e^{-(2k-1)^2\pi^2\tau} \times \cos[(2k-1)\pi\xi] \quad [21]$$

This relation can be used by a computer program to find the concentration of lithium salt at any time and position. Because this solution converges more quickly at long times, it is also useful to calculate a short-time solution, which we get by expanding the Laplace transform for large s and inverting

$$\theta = 2\sqrt{\tau}\left\{ \text{ierfc}\left(\frac{\xi}{2\sqrt{\tau}}\right) + \sum_{k=1}^{\infty} (-1)^k \left[\text{ierfc}\left(\frac{k+\xi}{2\sqrt{\tau}}\right) + \text{ierfc}\left(\frac{k-\xi}{2\sqrt{\tau}}\right)\right]\right\} \quad [22]$$

where ierfc is the integrated error function complement

$$\text{ierfc}(x) = \int_x^{\infty} \text{erfc}(x)dx = \frac{1}{\sqrt{\pi}}\left(e^{-x^2} - 2x\int_x^{\infty} e^{-x^2}dx\right) \quad [23]$$

Figure 2 shows the concentration profile yielded by a computer program employing Eq. 14, 21, and 22. In the computer program, values for θ were given by Eq. 22 if $\tau < 0.1$, Eq. 21 if $0.1 \leq \tau \leq 3$, and Eq. 14 if $\tau > 3$. The range of t has been chosen to span all regimes of the dimensionless time; the contour lines represent increments of 1 in ln(t). The limiting current density (i_L) in this example is 1.378 mA/cm^2; in Fig. 2, the current was set to 99.99% of i_L. Properties in this analysis were constant with respect to concentration; thus

$$i_L = \frac{2c_b DF}{(1-t_+^0)L}$$

Values for all transport properties were based on the oxymethylene-linked poly(ethylene oxide) (PEMO)-LiTFSI system at 85°C[19]: 5×10^{-12} m^2/s for D and 0.3 for t_+^0. The interelectrode spacing (L) was set to 100 μm. Note that under these conditions it took about 2,000 s for the concentration profile to reach steady state.

Potential profile.—The cell potential can be calculated using the results of the section on Concentration profile. To solve for potential in the solution relative to the cathode requires knowledge of the cathodic surface overpotential. We assumed that a Butler-Volmer type expression could be used for the reaction kinetics, and that the concentration of the salt affected only the cathodic rate, giving

$$\frac{i}{F} = k_a \exp\left(\frac{\alpha_a F}{RT}\Phi\right) - k_c c \exp\left(\frac{-\alpha_c F}{RT}\Phi\right) \quad [24]$$

where k_a and k_c are anodic and cathodic rate constants, R is the ideal gas constant, and T is the absolute temperature of the cell. The α_i are transfer coefficients. We assume that the potential relative to the cathode can be expressed as the sum of an open-circuit potential (OCP) and a surface overpotential

$$\Phi = V_0 + \eta_s \quad [25]$$

When the current is zero, the OCP is achieved; using Eq. 24, this yields the relationship

$$V_0 = \frac{RT}{(\alpha_a + \alpha_c)F}\ln\left(\frac{k_c c}{k_a}\right) \quad [26]$$

Assuming that the transfer coefficients sum to unity and inserting relations 26 and 25 into Eq. 24, we found that

$$\frac{i}{F} = k_a^{\alpha_c}(k_c c)^{\alpha_a}\left[\exp\left(\frac{\alpha_a F}{RT}\eta_s\right) - \exp\left(\frac{-\alpha_c F}{RT}\eta_s\right)\right] \quad [27]$$

Equation 27 allows definition of an exchange current density in terms of the kinetic parameters

$$i_0 = Fk_a^{\alpha_c}(k_c c)^{\alpha_a} = i_{0,\text{ref}}\left(\frac{c}{c_{\text{ref}}}\right)^{\alpha_a} \quad [28]$$

Assuming that $\alpha_a = \alpha_c = 1/2$ yields an analytic expression for the surface overpotential in terms of concentration

$$\eta_s = \frac{2RT}{F}\sinh^{-1}\left[\frac{1}{2}\left(\frac{i}{i_{0,\text{ref}}}\right)\left(\frac{c_{\text{ref}}}{c}\right)^{1/2}\right] \quad [29]$$

Equation 27 yields a boundary condition for the potential at the surface of the cathode. We could numerically solve for the surface overpotential if the symmetry factors were not equal, but because literature values are around 0.5 [20] we assume that Eq. 29 reasonably describes the system. With the overpotential as a boundary condition, we use the integrated form of modified Ohm's law[21] to find the instantaneous potential, Φ_3, at any point across the cell

$$\Phi_3(y,t) = \eta_s - \frac{i}{\Lambda}\int_0^y \frac{1}{c}dy + \frac{RT}{F}(1-2t_+^0)\ln\left[\frac{c(y,t)}{c(0,t)}\right] \quad [30]$$

The conductivity of the separator has been assumed to be linear in concentration; Λ is the equivalent conductance of the electrolyte. Assuming that the electrolyte is relatively dilute, we express Λ using the Nernst-Einstein relation

$$\Lambda = \frac{F^2 D}{2t_+^0(1-t_+^0)RT} \quad [31]$$

Figure 3 shows the potential in the separator (Φ_3), solved as a function of time, position, i, L, c_b, and T, and the properties D, t_+^0, $i_{0,\text{ref}}$, and c_{ref}. The figure shows simulations where $i_{0,\text{ref}}$ is set to 3 mA/cm^2; c_{ref} is assumed to be 1 M.[20] All other parameters were the same as those used for the concentration profile shown in Fig. 2. The long-time solution is not shown on the graph; it is dominated by the concentration and surface overpotentials, nearly immediately rising

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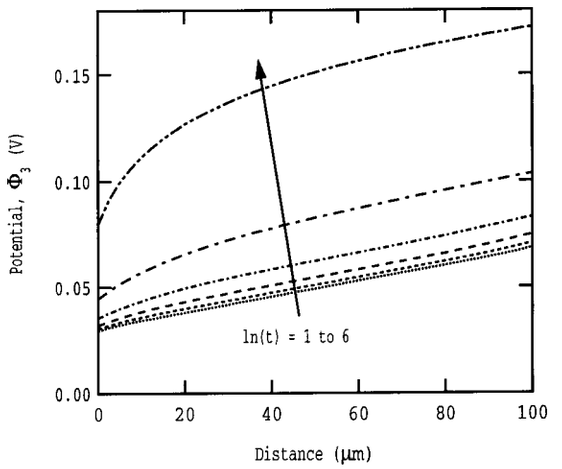


Figure 3. Potential *vs.* distance and time in the cell.

to 0.4 V. Contour lines on the graph represent intervals of 1 in ln(t). The potential became increasingly positive across the cell; this was expected, because the cathode, which was at the lowest potential ($\Phi_1 = 0$), has been placed at zero distance. Zero distance is a reasonable choice for position if we presume that roughness of the electrode is on a very small length scale (*i.e.*, much less than L). Note that the SEI resistance has been neglected in the potential calculations for simplicity.

Dendrite growth rate.—Now let us imagine that there is a small single dendrite protruding from the cathode. Following the method of Barton and Bockris,[12] we assume that mass transfer and surface forces dominate the kinetics of dendrite growth. In order to apply the theory in conditions where concentration and potential vary, we need to add well-defined thermodynamic reference points to the analysis. We can sketch the phases according to the diagram in Fig. 4. In the drawing, phase α is a reference electrode of a given kind, placed in the vicinity of the dendrite tip, *i.e.*, near the dendrite, but far enough away that the potential and concentration are the macroscopic transient values calculated in the sections on Concentration profile and Potential profile. It should be noted that the former work by Barton and Bockris and Diggle *et al.*[12,13] does not allow for changing concentration and potential in the vicinity of the dendrite tip as it grows.

We have implicitly assumed that there was only one dendrite, or, if there were more than one, that the dendrites were sufficiently far apart that they did not affect each other. Also, we have assumed that the dendrite was small enough that its presence did not greatly affect the concentration and potential profiles in the cell, and that its growth did not appreciably lower the bulk concentration of lithium. This is a good approximation for needle-like dendrites.[13] From the theory of liquid droplets,[22] we know that surface forces at the dendrite tip cause the pressure inside it to rise as its principal radii of curvature (r_1 and r_2) decrease. If we assume that the dendrite tip is hemispherical, the two radii of curvature are the same. This simplifies the expression for the pressure difference to

$$\Delta p = \gamma\left(\frac{1}{r_1} + \frac{1}{r_2}\right) = \frac{2\gamma}{r} \qquad [32]$$

where p is the pressure, r is the radius of curvature of the dendrite tip, and γ is the interfacial tension between the polymer separator and lithium. We assume that γ is independent of salt concentration. Because the cell is isothermal, we can relate the pressure difference to the chemical potential difference (Δμ) of curved lithium relative to planar lithium

$$\Delta\mu = V\Delta p = \frac{2\gamma V}{r} \qquad [33]$$

Here V is the molar volume of solid lithium, and is assumed to be the same in the dendrite or the electrode.

We wished to develop a kinetic equation similar to Eq. 27. The driving force for reaction was the surface overpotential (η_s) at the dendrite/electrolyte interface. Using the potentials (Φ_i) indicated in Fig. 4, we write

$$\eta_s = \Phi_1 - \Phi_2 - (\Phi_1 - \Phi_2)_{eq} \qquad [34]$$

The equilibrium potential can be expressed using Eq. 33

$$F(\Phi_1 - \Phi_2)_{eq} = \mu_{e^-}^{\phi} - \mu_{e^-}^{\alpha'} = (\mu_{Li}^{\phi} - \mu_{Li^+}^{\delta}) - (\mu_{Li}^{\alpha'} - \mu_{Li^+}^{\delta}) = -\frac{2\gamma V}{r} \qquad [35]$$

To use the computed concentration and potential profiles in the vicinity of the dendrite tip, and to concur with the dendrite literature, we needed to rewrite the potential in the kinetic expression as $\eta = \Phi_1 - \Phi_3$. Then it was useful to find

$$F(\Phi_2 - \Phi_3) = \mu_{e^-}^{\alpha} - \mu_{e^-}^{\phi} = RT\ln\left(\frac{a_{Li}^{\alpha}}{a_{Li}^{\phi}}\right) + RT\ln\left(\frac{c_{Li^+}^{\delta}}{c_{Li^+}^{\delta'}}\right) + F(\Phi^{\delta'} - \Phi^{\delta}) \qquad [36]$$

where a_{Li} are activities of lithium. We can drop the last term if we assume the liquid-junction potential across the transition region is small, and we can drop the first if phases ϕ and α are reference electrodes of a given kind. Using Eq. 34-36 with the definition of η shows the driving force for the reaction to be

$$\eta_s = \eta + \frac{2\gamma V}{rF} + \frac{RT}{F}\ln\left(\frac{c_{Li^+}^{\delta'}}{c_{Li^+}^{\delta}}\right) \qquad [37]$$

The kinetics were assumed to follow a Butler-Volmer type equation much like Eq. 27. However, a term was added so that surface curvature affected the reaction rate

$$i_n = i_{0,ref}\left(\frac{c_{Li^+}^{\delta}}{c_{ref}}\right)^{\alpha_a}\left[x_{Li}^{\alpha'}\exp\left(\frac{2\gamma V}{rRT}\right)\right]^{\alpha_c}\left[\exp\left(\frac{\alpha_a F\eta_s}{RT}\right) - \exp\left(\frac{-\alpha_c F\eta_s}{RT}\right)\right] \qquad [38]$$

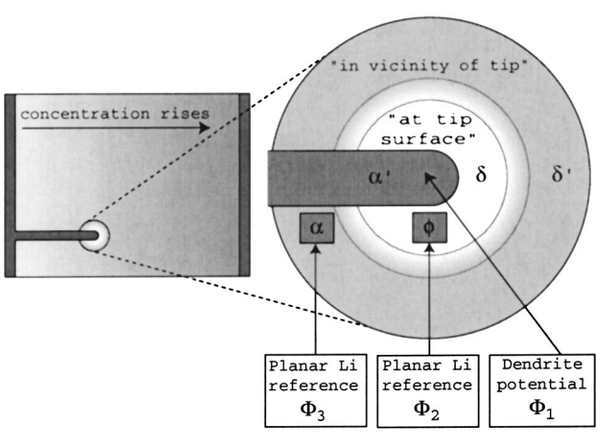


Figure 4. Diagram of the region near the dendrite tip.

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Equation 38 follows the development of Diggle *et al.*[13] a rigorous derivation is lengthy, and is the subject of present work by the authors. Note that Eq. 38 gives the current density normal to the (hemispherical) dendrite tip. Inserting Eq. 37 and assuming that the transfer coefficients sum to unity, we find

$$i_n = i_{0,\mathrm{ref}}\left(\frac{c^{\delta'}_{\mathrm{Li}^+}}{c_{\mathrm{ref}}}\right)^{\alpha_a}\left[x^{\alpha'}_{\mathrm{Li}}\exp\left(\frac{2\gamma V}{rRT}\right)\exp\left(\frac{\alpha_a F\eta}{RT}\right) - \frac{c^{\delta}_{\mathrm{Li}^+}}{c^{\delta'}_{\mathrm{Li}^+}}\exp\left(\frac{-\alpha_c F\eta}{RT}\right)\right] \quad [39]$$

The curvature thus appears to increase the anodic rate.

In practice, the concentration in phase δ is unknown. If we examine mass-transfer limitations, we can eliminate the concentration at the dendrite tip. With a spherical diffusion limitation, we say

$$i_n = \frac{DF(c^{\delta}_{\mathrm{Li}^+} - c^{\delta'}_{\mathrm{Li}^+})}{(1 - t^0_+)r} \quad [40]$$

Use of an alloy electrode would reduce $x^{\alpha'}_{\mathrm{Li}}$, linearly reducing the anodic rate. If we assume that the dendrite is pure lithium, then we can set the mole fraction in Eq. 39 to unity; solving Eq. 40 for $c^{\delta}_{\mathrm{Li}^+}$ and plugging into Eq. 39 yields the final expression for current density at the dendrite tip

$$\frac{i_n}{i_{0,\mathrm{ref}}} = \frac{\exp\left(\frac{2\gamma V}{rRT}\right)\exp\left(\frac{\alpha_a F\eta}{RT}\right) - \exp\left(\frac{-\alpha_c F\eta}{RT}\right)}{\left(\frac{c_{\mathrm{ref}}}{c^{\delta'}_{\mathrm{Li}^+}}\right)^{\alpha_a} + \frac{(1 - t^0_+)ri_{0,\mathrm{ref}}}{FDc^{\delta'}_{\mathrm{Li}^+}}\exp\left(\frac{-\alpha_c F\eta}{RT}\right)} \quad [41]$$

Equation 41 differs from the expression used by Diggle *et al.* in that it contains a thermodynamic reference parameter, c_{ref}, which significantly changes the functional dependence of dendrite tip current on concentration.[13] Material, kinetic, and transport properties, as well as the concentration and potential, are determined by the experimental system and the position of the dendrite in the cell. This leaves the dendrite tip radius as a variable. To find a worst-case scenario for dendrite growth, we need only maximize the tip current with respect to the radius.

Now the solutions from the previous sections can be used to simulate dendrite growth under constant current. The current density at the dendrite tip is related to the propagation velocity of the dendrite (v_{tip}) by

$$v_{\mathrm{tip}} = \frac{i_n(c,\eta)V}{F} \quad [42]$$

The concentration and potential are determined by the position of the dendrite tip in the cell. Thus, Eq. 42 can be used to find the maximum velocity of the dendrite as a function of c and η given all the other system properties. It can be phrased as a first-order differential equation and initial condition

$$\frac{\partial y_{\mathrm{tip}}}{\partial t} = \frac{V}{F}i_n(c,\eta) = \frac{V}{F}i_n(y_{\mathrm{tip}},t) \quad [43]$$

Table I. Parameters used in the dendrite growth model.

D	5×10^{-12} m^2/s	γ	1.716 J/m^2
t^0_+	0.3	c_b	1 M = 1000 mol/m^3
T	85°C = 358.15 K	c_{ref}	1 M = 1000 mol/m^3
R	8.3143 J/[mol K]	L	100 μm
F	96487 C/mol	$i_{0,\mathrm{ref}}$	30 A/m^2
V	1.2998×10^{-5} m^3/mol	$i = 0.95i_L$	$1.9c_bDF/[(1 - t^0_+)L]$

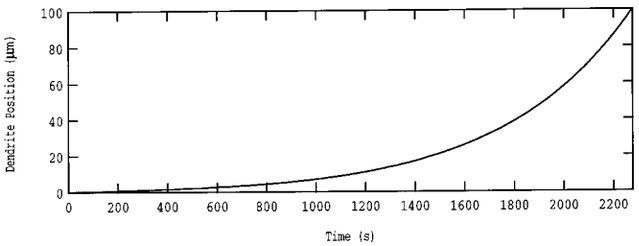


Figure 5. Typical growth profile for a lithium dendrite yielded by RK solution.

$$y_{\mathrm{tip}} = 0 \ \text{ at } \ t = 0 \quad [44]$$

Note that the initial condition 44 does not imply that the dendrite begins at a flat electrode, because growth is maximized with respect to tip radius at all times. Equation 43 was solved by a computer program using the Runge-Kutta (RK) solution method. For this solution, a surface energy of 1.716 J/m^2 was used, consistent with literature values for solid/polymer interfaces.[14] The RK time-step (h) was 0.5 s; because error in this solution goes as h^4, we expect that the solution is sufficiently precise. The other parameters are set to values used in the previous sections, summarized in Table I.

Simulation Results

The appearance of a dendritic growth profile is of primary interest in this analysis. Figure 5 shows a typical profile yielded by the simulation. The growth rate with respect to time and position is shown in Fig. 6. Figure 6 reveals that the dendrite accelerated both as time passed and as it moved across the cell. However, the shapes of these curves could not be easily described with simple curve fits; thus, the effect of changing variables on the qualitative shape of the growth curves was examined.

Adjusting the current density (expressed as the fraction of the limiting current density, $x\ i_L$) changed the relative impact of the transient concentration and potential profiles given in Eq. 14, 21, and 22. For this reason, the current density was varied to explore the shape of the dendrite growth profile. Representative shapes are shown in Fig. 7. Each of the profiles shown in the figure had similar character; they were all concave up, representing acceleration of the dendrite across the cell. At early times, the growth of the dendrite was relatively linear, corresponding to a slope of 1 in the graph. Lowering the current density appeared to prolong this relatively linear growth regime.

Experimental parameters were varied to evaluate their effect on t_s. For a better qualitative metric, we used Q_s, the charge passed per unit area before the cell shorts out. To fix the separator material in a cell sets the temperature and material properties, leaving two primary experimental variables: the current density and the gap width in the cell. The effect of current density on the charge passed at failure was evaluated, leaving all other parameters set to the values shown in Table I. The result is shown in Fig. 8, which also

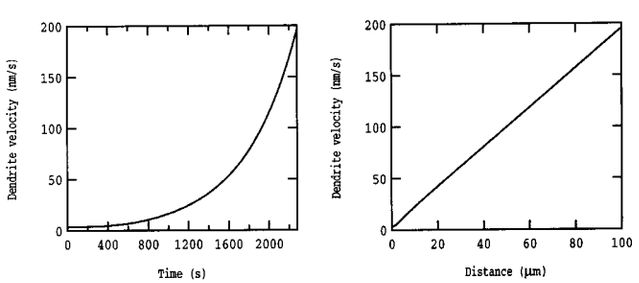


Figure 6. Dendrite velocity as a function of time and distance.

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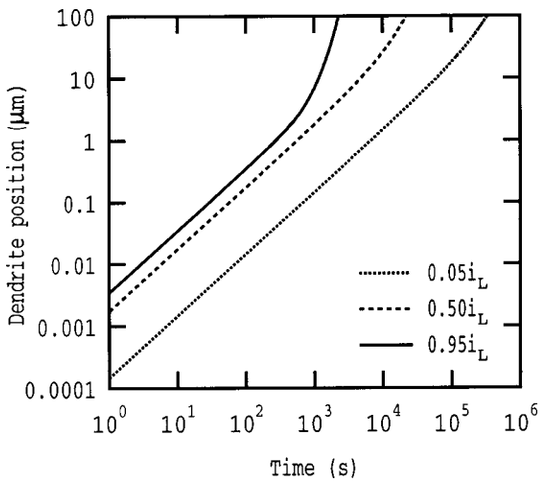


Figure 7. Dendrite growth profile shape at varying currents.

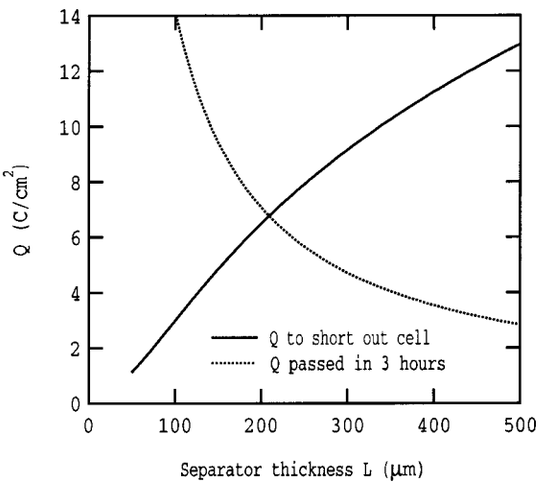


Figure 9. Charge to short out cell and charge in 3 h *vs.* interelectrode distance. $i = 0.95^{*}i_{L}$.

shows the charge passed in three hours; if the cell-shortage line lies to the right of the three-hour line, failure occurs during a single typical charge cycle.

Figure 8 is of interest for two main reasons. The limiting current clearly must not be approached under normal operating conditions; however, Fig. 8 suggests that the cell fails if run above 75% of the limiting current, placing a much more conservative limit on the charge rate. In addition, although the concavity of the Q_s curve changed, the slope is negative for all values of current, indicating that lowering the current density always increases the charge that can be passed before failure.

The change in Q_s with respect to gap width is shown in Fig. 9, where the cell was simulated at $0.95i_L$. In this figure, if the cell-shorting line lies to the left of the 3 h line, failure occurs during a single charge. Q_s consistently increases with increasing interelectrode distance. However, the curve is concave down, indicating that the benefits of increased separator thickness decrease as the cell gets thicker. This is because the dendrite accelerates as potential difference rises; Fig. 3 clearly shows that the potential drop only increases with L. It should be borne in mind that the resistance of the dendrite has been neglected in our analysis; a potential difference between the dendrite tip and base would decrease this acceleration effect. Taking the dendrite resistivity to be that of bulk lithium metal,[23] we found that the potential drop in a 100 μm dendrite would be 116 pV, justifying its neglect.

Now let us explore the effects of changing the separator material. Such a change would vary any of three experimental parameters: γ, D, and t_+^0. The surface energy is related to the adhesion of the polymer to the electrode and the character of the SEI; the other two properties relate to polymer and salt structure.

The surface energy was varied to evaluate its effect on Q_s, shown in Fig. 10. Although surface energy is a basic factor in the dendrite growth kinetics, a rise over three orders of magnitude in γ increases Q_s by only ~6%, indicating that surface forces do not strongly control the growth rate of dendrites under the conditions in question. This implies that a more adhesive polymer, a surface coating on the electrode, or a stiffer SEI will not significantly slow dendrite propagation. Note that this analysis assumes the SEI to have zero thickness or mechanical strength, its properties wholly reflected by an isotropic surface energy term. Though γ is assumed isotropic here, anisotropy may affect the shapes of propagating dendrites; we assume that the minimal impact of γ on Q_s justifies neglect of this consideration.

Because we have used the Nernst-Einstein relationship to reach Eq. 31, the diffusion coefficient and transference number are inti-

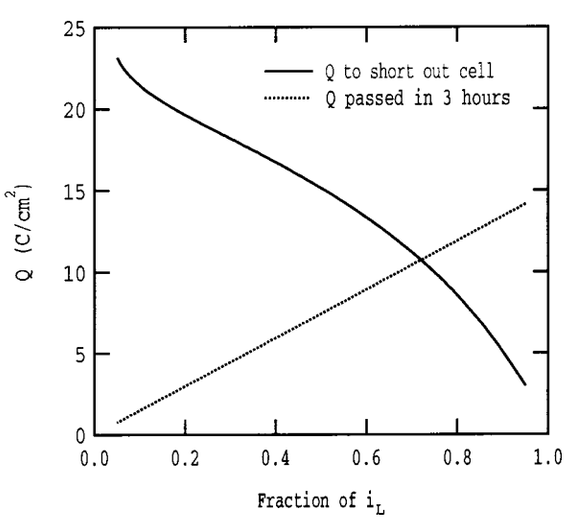


Figure 8. Charge to short out cell and charge in 3 h *vs.* fraction of i_L.

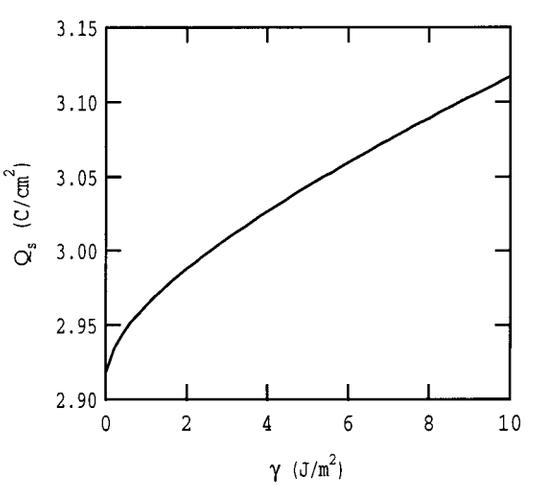


Figure 10. Effect of changing γ on Q_s.

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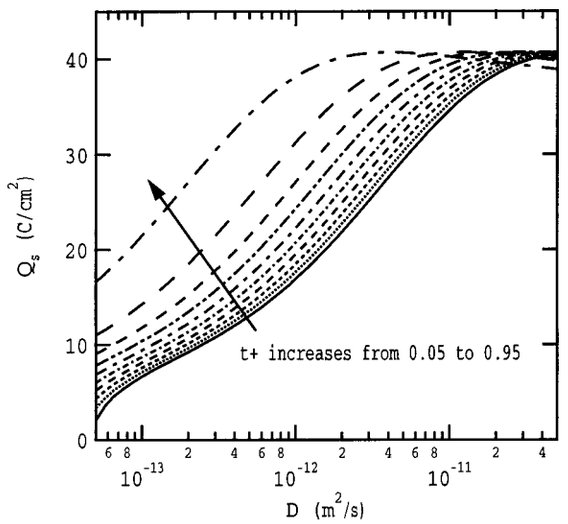


Figure 11. Effect of changing D and t_+^0 on Q_s.

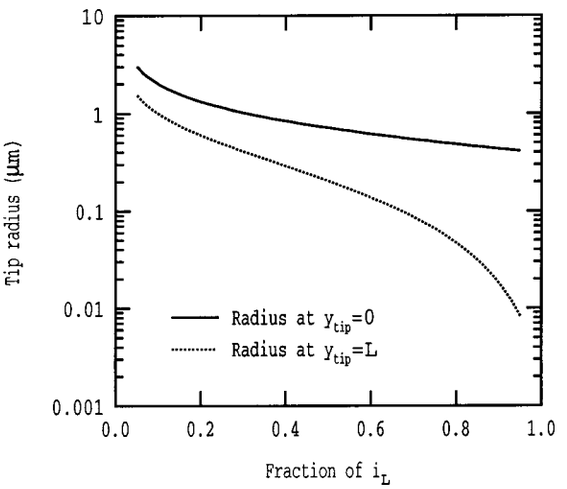


Figure 12. Initial and final dendrite tip radius *vs.* fraction of the limiting current.

mately related to the conductivity in the growth model. For this reason, data sets were developed varying both of these variables. A constant current density of 96.5 mA/cm^2, $0.95i_L$ at the lowest D and t_+^0 values, was used. The result of this simulation is shown in Fig. 11, where contour lines represent intervals of 0.10 in t_+^0. In the region shown, the curve shows maxima at the upper contours, indicating that there are combinations of D and t_+^0 which allow the charge passed before failure to be maximized. From Eq. 31 we see that the equivalent conductance is linear with respect to diffusion coefficient; the changing concavity of the curves thus indicates that Q_s cannot be predictably altered by increasing or decreasing the electrolyte conductivity, especially at high transference numbers. Though unity t_+^0 is an ideal situation for a salt, Eq. 2 and 31 prevent choosing it as a condition for simulation.

Impact of Assumptions

We have gathered some useful results regarding the effect of various system parameters on dendrite growth. Now we explore the assumptions made in the development of the simulation and question their validity. The most significant assumptions are listed in Table II. Many of these assumptions are standard for systems containing dendrites, as addressed in the introduction. However, there are a few that remain questionable in the lithium/polymer system: specifically, assumptions 2-4, 6-8, and 17. These assumptions provide excellent suggestions for future work on the lithium/polymer dendrite problem.

Table II. Assumptions used in development of the dendrite growth model.

1. One-dimensional
2. One dendrite (or broad spacing)
3. Dendrite is needle-like
4. D, t_+^0, and γ constant with respect to concentration
5. Isothermal
6. Ideal solution (dilute)
7. Constant density
8. Λ found by Nernst-Einstein relationship
9. Electroneutrality holds throughout the cell
10. Anion controls migration of the salt
11. Concentration of salt affects cathodic rate
12. Curvature of tip affects anodic rate
13. Tip is hemispherical
14. Spherical diffusion limitation
15. No liquid-junction potential near tip
16. Dendrite has no resistance
17. Tip is static (no viscous or mechanical force)
18. Surface tension is isotropic
19. Growth rate is maximized with respect to tip radius

Chazalviel has performed experiments which suggest well-spaced needle-like dendrites, consistent with assumption 2. However, because this work was performed over the limiting current, it may not apply to cycling of secondary batteries.[17] Though theoretical work by Yamaki *et al.* showed that the dendrites were needle-like, the lithium electrode is treated as a Newtonian fluid, which may not be a reasonable assumption relative to a polymer interface. In addition, the interfacial tension was used as a bulk pressure which drives flow in the lithium, an inappropriate treatment of this force.[18] Therefore, assumption 3 is also not strongly supported by the literature. In order to rectify these difficulties, simple microscopy experiments may be necessary to observe lithium dendrite growth directly.

In work for his doctoral dissertation, Hafezi[24] has shown that transport properties vary strongly with respect to concentration and that the PEMO/LiTFSI system may significantly deviate from ideality. For this polymer/salt combination, the variation of density has not been well-established, but Brissot, *et al.* found that density in poly(ethylene oxide)/LiTFSI (a similar system) changes somewhat with concentration.[25] Assumptions 4, 6, 7, and 8 thus require inspection. Since data have recently been made available showing the concentration dependence of the diffusion coefficient, transference number, and conductivity in this system, it would be useful to modify the simulation developed here to allow variable transport properties. Thermodynamic factors have also been presented by Hafezi; therefore, the assumptions regarding reaction kinetics (11-15) may need to be modified to rely on salt activity rather than directly on concentration. Some further experiments may be necessary to measure the concentration dependence of polymer density, or at least to verify the validity of 7.

Assumption 19 is not listed as questionable because maximizing rate provides a worst-case scenario for dendrite growth. Simulation results for tip radius are given in Fig. 12, which shows initial and final tip radii at varying current densities. The figure shows that initial tip radii were of order ~1 μm at all currents; final tip radii were of order ~1 μm at low currents but dropped to ~10 nm at higher currents. Though the fractal analysis presented by Chen and Jorné[7] suggests that at its root the problem of dendrite growth is statistical in nature, the tip radii shown in Fig. 12 are much larger than the atomic radius of lithium (1.57 Å); we can conclude that the continuum hypothesis is valid for this analysis. Also, it should be noted that the tip radius is the distance over which diffusion occurs;

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thus electroneutrality is probably not violated according to the hypothesis of Chazalviel.[17]

The last significant questionable assumption is 17, which suggests the most important direction for future theoretical work. The theory developed by Barton and Bockris, like all of the work on surface instability discussed in the introduction to this paper, presumes that surface forces lead directly to dendrite growth.[12] However, neglect of viscous and elastic forces in the lithium/polymer system means that the improved dendrite resistance shown to exist in some common experimental systems cannot be explained theoretically. These systems include those with increased pressure on or electrolyte flow between electrodes[4,10] and those containing gelled or mechanically strengthened polymers.[11]

Conclusions

A model has been developed to describe dendrite growth in a parallel-electrode lithium/polymer cell under galvanostatic charging with transient concentration and potential profiles. The Barton and Bockris theory of dendrite growth kinetics has been corrected to incorporate well-defined thermodynamic reference points. Using data for the PEMO/LiTFSI system, we found that dendrites always accelerate across cells, and that growth can always be slowed by lowering the current density. Cell shorting occurred during typical charges above 75% of i_L. Interelectrode distance can be increased to slow cell failure, but the advantages decrease as distance lengthens. Surface forces were shown to have little effect on cell failure, and no consistent trend was observed when varying the transport properties in the separator.

The model could be improved in several ways. Some experiments may be necessary to characterize polymer properties more thoroughly with respect to salt concentration, and variable properties could be incorporated into a computer program to model growth more realistically. Variable properties have ramifications not only with regard to potential and concentration profiles in the cell, but also the kinetics of the reaction, which may be affected by use of chemical activity rather than concentration of the salt. Usage of the continuum hypothesis may belie the statistical underpinnings of the deposition process. Therefore, we must assume that this analysis is highly idealized and may not precisely predict cell failure in actual systems. The magnitude of the uncertainty is not clear.

Several avenues are thus opened for further work. A variable-property simulation should be performed to describe the system more accurately; recent experimental data make this work possible. The relative magnitude of bulk mechanical and hydrodynamic forces also needs to be evaluated; this would provide insight toward the assumption that surface forces determine the rate of dendritic propagation.

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List of Symbols

a_{Li}	activity of lithium, unitless
A	electrode area, m^2
B	integration constant
c	salt concentration, mol/m^3
c_b	initial bulk concentration of lithium salt, mol/m^3
c_{ref}	reference concentration of lithium salt, mol/m^3
C	integration constant
D	salt diffusion coefficient, m^2/s
F	Faraday's constant, 96487 C/mol
h	Runge-Kutta time step
i	current density, A/m^2
i_n	current density normal to dendrite tip, A/m^2
i_L	$= 2c_bDF/(1 - t_+^0)L$ limiting current density, A/m^2
I	current, A
k_a	anodic rate constant, mol/s
k_a	cathodic rate constant, m^3/s
L	interelectrode distance, m
LiTFSI	lithium bis-(trifluoromethane sulfonyl)-imide $[Li(CF_3SO_2)N]$
p	pressure, Pa
PEMO	oxymethylene-linked polyethylene oxide ($[(CH_2CH_2O)_{8.7} - CH_2O]$, Mw = 100 kg/mol)
Q	charge passed per unit area, C/cm^2
Q_s	charge passed per unit area before cell failure, C/cm^2
R	ideal gas constant, 8.3143 J/mol K
r	dendrite tip radius, m
RK	Runge-Kutta
SEI	solid-electrolyte interphase
t	time, s
t_s	time passed before cell failure, s
t_+^0	cation transference number, unitless
T	absolute temperature, K
V	molar volume of lithium, 1.2998×10^{-5} m^3/mol
V_0	open-circuit potential, V
v_{tip}	dendrite propagation velocity, m/s
x_{Li}	mole fraction of lithium, unitless
y_{tip}	dendrite tip position, m

Greek

α_a	anodic transfer coefficient (unitless)
α_c	cathodic transfer coefficient, unitless
γ	surface energy, J/m^2
$\eta = \Phi_1 - \Phi_3$	dendrite kinetic potential, V
η_s	surface overpotential, V
$\theta = (c - c_b)DF/(1 - t_+^0)iL$	dimensionless concentration
θ_{SS}	steady-state dimensionless concentration
θ_T	transient dimensionless concentration
κ	electrolyte conductivity, S/m
$\Lambda = F^2D/t_+^0(1 - t_+^0)RT$	equivalent conductance, Sm^2/mol
μ	chemical potential, J/mol
$\xi = y/L$	dimensionless distance
$\tau = Dt/L^2$	dimensionless time
Φ	quasielectrostatic potential, V
Φ_1	potential at cathode or dendrite tip, V (see Fig. 4)
Φ_2	potential at dendrite tip surface, V (see Fig. 4)
Φ_3	potential in vicinity of dendrite tip, V (see Fig. 4)

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